

Explain and check long multiplication

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Miro calculates 326×45 as 1,467. Identify the likely place-value error. Copy or complete the first step.

2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

3. Calculate and check $6,014 \times 38$.

4. Miro says: Miro trusts a written answer without estimating first. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Miro calculates 326×45 as 1,467. Identify the likely place-value error. Copy or complete the first step.

Answer He treated 40 as 4. The correct product is 14,670.

2. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

Answer 120,000.

3. Calculate and check $6,014 \times 38$.

Answer 228,532.

4. Miro says: Miro trusts a written answer without estimating first. Is this correct? Explain.

Answer An estimate reveals place-value errors before the exact calculation is accepted.

Explain and check long multiplication

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

2. Calculate and check $6,014 \times 38$.

3. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.

4. Identify and correct this misconception: Miro trusts a written answer without estimating first.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. Which is closer to $4,208 \times 29$: 120,000 or 1,200,000?

Answer 120,000.

2. Calculate and check $6,014 \times 38$.

Answer 228,532.

3. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.

Answer Answers vary, for example $5,000 \times 41 = 205,000$.

4. Identify and correct this misconception: Miro trusts a written answer without estimating first.

Answer An estimate reveals place-value errors before the exact calculation is accepted.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Explain and check long multiplication

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.

2. Explain why this reasoning fails: Miro trusts a written answer without estimating first.

3. Create a new example that tests the same idea as: Calculate and check $6,014 \times 38$.

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Create a four-digit by two-digit multiplication with a product between 200,000 and 210,000.
Answer Answers vary, for example $5,000 \times 41 = 205,000$.
2. Explain why this reasoning fails: Miro trusts a written answer without estimating first.
Answer An estimate reveals place-value errors before the exact calculation is accepted.
3. Create a new example that tests the same idea as: Calculate and check $6,014 \times 38$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.